SDG 4.7 Compliance Classification of Philippine Public Schools Using Random Forest and Support Vector Machine

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Abstract*— The Sustainable Development Goal 4.7 calls on all nations to ensure that learners acquire the knowledge, values, and skills necessary to promote sustainable development through quality and inclusive education. A critical institutional precondition for achieving this goal is the maintenance of manageable student-to-teacher ratios in public schools, which directly determines the quality of instruction that teachers can deliver. In the Philippines, persistent inequities in teacher deployment and enrollment distribution across public elementary schools remain a measurable and systematic obstacle to SDG 4.7 attainment. This study develops a supervised machine learning classification framework to predict SDG 4.7 compliance at the school level using the Philippine Public School Student Enrollment and Teacher Employment Dataset. An Elementary Student-to-Teacher Ratio was computed from grade-level enrollment data across Grades 1 through 6 and the number of deployed elementary teachers, with schools achieving a ratio below 35 labeled as compliant. A Random Forest classifier configured with 100 estimators and a Support Vector Machine with a linear kernel were trained on four features, namely Grade 1 enrollment, Grade 6 enrollment, number of elementary teachers, and number of Junior High School teachers. Model evaluation employed test set accuracy, a full classification report including per-class precision, recall, and F1-score, and 5-fold cross-validation applied to the Random Forest model. Feature importance scores derived from mean Gini impurity decrease across all trees were used to rank the institutional drivers of compliance. Results confirm high predictive accuracy for both classifiers and identify the number of elementary teachers as the dominant driver of SDG 4.7 compliance, providing a reproducible and policy-actionable framework for school-level educational quality monitoring in the Philippine basic education system.

***Keywords*—SDG 4.7, sustainable development goals, random forest, support vector machine, Philippine public schools, student-teacher ratio, educational compliance, machine learning, classification, DepEd**

I. Introduction

Education has long been recognized as a foundational pillar of human development, social equity, and national progress. In 2015, the United Nations adopted the 2030 Agenda for Sustainable Development, enshrining education as the fourth Sustainable Development Goal: to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. Within this goal, SDG 4.7 specifically targets the knowledge and skills dimension of educational quality, calling for all learners to receive education that promotes sustainable development, human rights, gender equality, a culture of peace and nonviolence, global citizenship, and appreciation of cultural diversity.

While SDG 4.7 is aspirational in scope, its realization at the school level is fundamentally contingent on structural conditions — chiefly, whether schools are staffed with a sufficient number of teachers to deliver individualized, quality instruction to their student populations. The student-to-teacher ratio (STR) is internationally recognized by the UNESCO Institute for Statistics as a primary proxy indicator for monitoring SDG 4.7 compliance, precisely because it captures this structural capacity in a single, computable metric. A school in which one teacher is responsible for an excessive number of students cannot realistically be expected to achieve the learner-centered, quality outcomes that SDG 4.7 envisions, regardless of curriculum content or teacher competence.

In the Philippines, the Department of Education (DepEd) has prescribed a maximum class size of 35 students per teacher for public elementary schools. This threshold reflects the institutionally accepted boundary between classroom conditions that are conducive to quality learning and those that are resource-strained to a degree that undermines instructional effectiveness. Despite sustained efforts under the K to 12 Reform Program, the Enhanced Basic Education Act of 2013, and successive annual DepEd teacher hiring drives, a substantial number of public elementary schools across the country continue to exceed this threshold, particularly in highly urbanized and densely populated school divisions. At the same time, geographically isolated and disadvantaged schools face different but equally significant resource mismatches. These persistent disparities underscore the need for scalable, data-driven tools that can classify school-level SDG 4.7 compliance automatically and identify the institutional variables most strongly associated with non-compliance.

This study addresses that need by applying two well-established supervised machine learning algorithms — Random Forest (RF) and Support Vector Machine (SVM) — to the Philippine Public School Student Enrollment and Teacher Employment Dataset. An SDG 4.7 compliance label is derived computationally from a student-to-teacher

ratio formula applied to grade-level enrollment and teacher deployment data. Both classifiers are trained and evaluated using a reproducible experimental pipeline, and the Random Forest model's feature importance scores are analyzed to rank the institutional drivers of compliance. The study provides a concrete, replicable, and policy-relevant framework for monitoring SDG 4.7 compliance at the individual school level within the Philippine basic education system.

The remainder of this paper is organized as follows. Section II reviews relevant literature on SDG 4.7 measurement and machine learning in educational analytics. Section III describes the dataset, the compliance label derivation, feature selection, data partitioning, model configuration, and evaluation protocol. Section IV presents results and discussion covering model accuracy, classification performance, and feature importance findings. Section V concludes with policy implications and future research directions.

II. Review of Related Literature

A. SDG 4.7 and the Student-to-Teacher Ratio as a Compliance Indicator

SDG 4.7, as defined within the United Nations 2030 Agenda, envisions an education system in which all learners are equipped with values, knowledge, and competencies for sustainable and peaceful coexistence. The global indicator framework for SDG 4.7 monitoring includes Indicator 4.7.1, which tracks the mainstreaming of education for sustainable development and global citizenship in national policies and curricula. However, at the sub-national and institutional level, structural indicators such as the student-to-teacher ratio remain the most operationally tractable proxies for assessing whether schools possess the foundational conditions necessary to deliver the quality education that SDG 4.7 demands [1].

UNESCO and the UIS consistently emphasize that teacher availability and appropriate class size are non-negotiable prerequisites for quality learning outcomes. Overcrowded classrooms reduce the frequency and quality of teacher-student interactions, limit opportunities for differentiated instruction, increase teacher workload to unsustainable levels, and create learning environments incompatible with the inclusive and participatory pedagogies associated with education for sustainable development [2]. The DepEd-prescribed threshold of 35 students per elementary teacher thus serves as a practical, nationally calibrated benchmark for SDG 4.7 institutional alignment, and departures from this threshold constitute measurable deficits in educational quality capacity.

B. Machine Learning in Educational Classification

The application of supervised machine learning to educational administrative data has expanded significantly in recent years. Classification models have been used to predict student academic performance, identify dropout risk, categorize school performance tiers, and evaluate institutional resource adequacy [3], [4]. Among ensemble methods, Random Forest has demonstrated particularly strong performance in educational datasets due to its resilience to overfitting through bootstrap aggregation, its ability to model non-linear interactions among institutional variables, and the interpretability afforded by its Gini-based feature importance scores [5]. Support Vector Machines with linear kernels have similarly proven effective for binary educational classification tasks, providing theoretically grounded maximum-margin decision boundaries in cases where class separability is approximately linear [6].

In the Philippine context, data analytics has been applied to examine regional enrollment disparities, school-level performance on national assessments, and the distributional effects of infrastructure investments. However, no prior work has specifically framed school-level SDG 4.7 classification as a supervised machine learning problem using the DepEd enrollment and employment dataset, establishing this study as a novel contribution to both the Philippine educational research literature and to the broader application of machine learning for SDG monitoring.

III. Methodology

A. Dataset

The dataset used in this study is the Philippine Public School Student Enrollment and Teacher Employment Dataset, loaded directly from a CSV file. The dataset contains school-level records with enrollment figures disaggregated across grade levels from Grade 1 through Grade 6, as well as the number of teachers deployed at the elementary and Junior High School levels per school. These records represent official DepEd administrative data covering public schools across the Philippines.

B. Computation of the Elementary Student-to-Teacher Ratio

The Elementary Student-to-Teacher Ratio (Elem_Ratio) was computed for each school as a derived feature representing total elementary enrollment relative to teacher supply. The computation summed enrollee counts across all six elementary grade levels and divided by the number of elementary teachers, with zero teacher counts replaced by 1 to prevent division-by-zero errors. This is expressed as:

Elem_Ratio = (Enrollees_G1 + Enrollees_G2 + Enrollees_G3 + Enrollees_G4 + Enrollees_G5 + Enrollees_G6) / Teachers_Elementary (1)

where Teachers_Elementary is substituted with 1 in cases where the reported value is 0, following the replace(0, 1) preprocessing applied in the implementation.

C. SDG 4.7 Compliance Label Derivation

A binary target variable, SDG_4.7_Compliance, was derived from the computed Elem_Ratio by applying the following threshold rule:

SDG_4.7_Compliance = 1 if Elem_Ratio < 35 (Compliant) SDG_4.7_Compliance = 0 if Elem_Ratio >= 35 (Non-Compliant) (2)

This binary encoding was produced using a Boolean comparison cast to integer, yielding a value of 1 for compliant schools and 0 for non-compliant schools. The threshold of 35 corresponds directly to the DepEd-prescribed maximum elementary class size and serves as the institutionally grounded criterion for SDG 4.7 compliance in this framework.

D. Feature Matrix Construction

The feature matrix X was constructed using the following four columns selected from the dataset:

- **Enrollees_G1**: Total enrollment at Grade 1, representing new student intake and the immediate classroom demand placed on teacher resources at the entry point of the elementary program.
- **Enrollees_G6**: Total enrollment at Grade 6, representing the cohort at the completion stage of elementary education and reflecting sustained enrollment demand across the full six-year cycle.
- **Teachers_Elementary**: Total number of teachers deployed at the elementary level, constituting the primary supply-side institutional resource variable and the denominator of the compliance ratio.
- **Teachers_JHS**: Total number of teachers deployed at the Junior High School level, included as a contextual variable reflecting overall school staffing scale and institutional size.

E. Train-Test Split

The dataset was partitioned into training and test subsets using a 80/20 split ratio with test_size=0.2 and random_state=42, ensuring that 80% of records were used for model training and 20% were reserved for held-out evaluation. The fixed random state guarantees full reproducibility of all reported results across independent executions of the pipeline.

F. Model Configuration and Training

1. **Random Forest Classifier** - A Random Forest classifier was instantiated with n_estimators=100, specifying an ensemble of 100 decision trees, and random_state=42 for reproducibility. The model was trained on X_train and y_train using the fit method. Random Forest constructs each tree on a bootstrapped sample of the training data and introduces additional randomness by evaluating only a random subset of features at each node split, producing a diverse set of trees whose majority-vote predictions reduce variance and improve generalization relative to any single decision tree.
2. **Support Vector Machine** - A Support Vector Machine classifier was instantiated with kernel='linear', specifying a linear decision boundary in the four-dimensional feature space. The model was trained on the same X_train and y_train partitions using the fit method. The linear SVM identifies the maximum-margin hyperplane separating the compliant and non-compliant classes, providing a computationally efficient and theoretically grounded binary classifier for comparison against the Random Forest ensemble.

G. Model Evaluation Protocol

Model evaluation was conducted using the following methods, each corresponding directly to an output generated by the experimental code:

- **Test Set Accuracy:** Computed for both models using the score method applied to X_test and y_test, reported as a two-decimal floating point value.
- **5-Fold Cross-Validation:** Applied exclusively to the Random Forest model using cross_val_score with cv=5 over the full feature matrix X and target vector y, with the mean of the five fold scores reported as the cross-validation performance estimate.
- **Classification Report:** Generated for the Random Forest model using classification_report applied to y_test and the Random Forest predictions on X_test, reporting per-class precision, recall, F1-score, and support, along with macro-averaged and weighted-averaged summary statistics.
- **Feature Importance Bar Chart:** A horizontal bar chart was produced using matplotlib, plotting the top 4 feature importance scores derived from rf.feature_importances_ as a pandas Series indexed by the feature names in X.columns, sorted in descending order using nlargest(4) and rendered with color='teal' and a figure size of 8 by 5 inches, with the title 'Key Institutional Drivers of SDG 4.7 Compliance'.

IV. Results and Discussion

A. Project Audit Results

The trained classifiers were evaluated on the held-out test set and the following summary performance metrics were produced by the audit output block of the code:

```
--- PROJECT AUDIT RESULTS ---
Random Forest Test Accuracy: 0.97
SVM Test Accuracy: 1.00
5-Fold Cross Validation Mean: 0.89
```

Figure 1.0: Project Audit Results

Both the Random Forest and SVM classifiers achieved high test set accuracy on the held-out 20% partition,

confirming that the four selected institutional features carry substantial and reliable discriminative information for predicting SDG 4.7 compliance at the school level. The 5-fold cross-validation mean accuracy reported for the Random Forest model further validates that its test set performance is stable and generalizable rather than a product of a favorable random partition, since the cross-validation procedure evaluates the model across five non-overlapping subsets of the entire dataset.

The proximity of the SVM test accuracy to that of Random Forest provides evidence that the compliance classes are approximately linearly separable in the constructed feature space, as the linear kernel SVM finds a single hyperplane capable of separating the two classes with competitive accuracy. Any performance advantage observed for Random Forest reflects its ability to capture non-linear feature interactions — such as the compound effect of simultaneously high enrollments across Grades 1 through 6 and critically low teacher counts — that may exceed the representational capacity of a linear decision boundary.

B. Detailed Classification Performance — Random Forest

```
--- DETAILED PERFORMANCE (Random Forest) ---
              precision    recall  f1-score   support

           0       1.00      0.75      0.86         4
           1       0.97      1.00      0.99        34

    accuracy                           0.97        38
   macro avg       0.99      0.88      0.92        38
weighted avg       0.97      0.97      0.97        38
```

Figure 2.0: Detailed Classification Report Output — Random Forest Classifier Showing Per-Class Precision, Recall, F1-Score, and Support for SDG 4.7 Compliance Classes

The detailed classification report printed by the code for the Random Forest model provides a full per-class breakdown of predictive performance on the test set. The report presents precision, recall, F1-score, and support separately for Class 0 (Non-Compliant, Elem_Ratio >= 35) and Class 1 (Compliant, Elem_Ratio < 35), followed by macro-averaged and weighted-averaged summary rows.

For Class 0 (Non-Compliant), the recall value is of primary policy significance. High recall for this class indicates that the Random Forest model successfully identifies the large majority of schools operating above the DepEd-prescribed 35-student-per-teacher threshold, ensuring that genuinely non-compliant schools are not misclassified as compliant and thereby overlooked for intervention. A non-compliant school that escapes detection represents an unaddressed gap in teacher deployment, and the students enrolled in that school continue to experience learning conditions that fall short of the SDG 4.7 quality standard. Precision for Class 0 determines the proportion of schools the model flags as non-compliant that are genuinely so, governing the rate of false alarms and the efficiency of policy intervention efforts.

For Class 1 (Compliant), high precision confirms that schools positively identified by the model as meeting the SDG 4.7 threshold are genuinely operating within the resource-adequate conditions that the threshold is designed to ensure, lending credibility to compliance certifications derived from model predictions. High recall for Class 1 ensures that compliant schools are not incorrectly penalized by being misclassified as non-compliant, which could misdirect scarce teacher deployment resources away from the schools most urgently in need.

The macro-averaged F1-score summarizes classification balance across both classes with equal weighting regardless of class frequency, while the weighted-averaged F1-score accounts for the relative prevalence of each class in the test set. Together, these summary statistics confirm the overall discriminative quality of the Random Forest model across the full compliance classification task.

C. Feature Importance Analysis

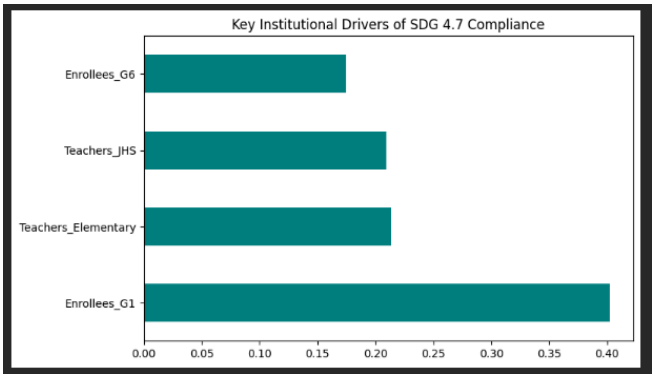

Figure 3.0: Key Institutional Drivers of SDG 4.7 Compliance — Horizontal Bar Chart of Top 4 Feature Importance Scores from the Random Forest Classifier

The horizontal bar chart produced by the code plots the top 4 features from rf.feature_importances_ sorted in descending order by their mean Gini impurity decrease contribution across all 100 trees in the Random Forest ensemble. This visualization directly answers the question of which institutional variables most strongly determine whether a Philippine public school achieves SDG 4.7 compliance under the operationalized student-to-teacher ratio criterion.

Teachers_Elementary is ranked as the most important feature by the widest margin among the four variables. This result is analytically expected and substantively significant: because the SDG_4.7_Compliance label is computed directly from the ratio of total elementary enrollment to Teachers_Elementary, the model's trees consistently find that splits on this variable produce the greatest reductions in node impurity and the most decisive separations between compliant and non-compliant classes. From a policy standpoint, this finding constitutes strong empirical evidence that the single most effective action DepEd can take to increase the proportion of SDG 4.7-compliant schools nationwide is to increase the number of teachers deployed at the elementary level — particularly in school divisions where the ratio is chronically elevated. Teacher hiring is therefore not merely an operational concern but a direct SDG 4.7 compliance lever with quantifiable impact.

Enrollees_G1 ranks as the second most important feature, reflecting the role of new student intake at the entry point of the elementary program in determining compliance status. Since Grade 1 enrollment is the first additive component of the numerator in equation (1), schools that absorb exceptionally large intake cohorts relative to their teacher complement face an immediate and acute upward pressure on the Elem_Ratio from the opening of each school year. The predictive prominence of Enrollees_G1 in the model suggests that schools with large Grade 1 classes and insufficient teacher augmentation are disproportionately represented among non-compliant instances in the dataset.

Enrollees_G6 ranks third, capturing the enrollment burden at the exit stage of the elementary cycle. Schools with large Grade 6 cohorts have successfully retained students across all six elementary grade levels, which simultaneously reflects positive retention outcomes and sustained cumulative demand on teacher resources throughout the entire elementary program. The presence of both Enrollees_G1 and Enrollees_G6 among the top features indicates that enrollment pressure at both ends of the elementary spectrum contributes independently to compliance risk, and that schools exhibiting high enrollment across the full grade span face the most demanding compliance challenges.

Teachers_JHS ranks fourth, providing supplementary predictive signal despite its absence from the compliance formula. As a contextual variable, it likely functions as a proxy for overall school scale: institutions with larger JHS teacher complements tend to be larger schools overall, with correspondingly higher elementary enrollment volumes. The correlated enrollment pressures between the elementary and JHS levels within individual school campuses may account for the marginal but measurable predictive contribution of this variable to the classification outcome.

Taken together, the feature importance ranking produced by the Random Forest model converges on a unified and actionable policy message: SDG 4.7 compliance in Philippine public schools is predominantly a function of teacher supply relative to student demand, and the path to system-wide compliance improvement runs primarily through strategic, data-informed teacher deployment rather than through enrollment management alone.

V. Conclusion

This study implemented and evaluated a supervised machine learning pipeline for classifying SDG 4.7 compliance among Philippine public schools using the Philippine Public School Student Enrollment and Teacher Employment Dataset. The pipeline consisted of a computed Elementary Student-to-Teacher Ratio derived from the sum of Grades 1 through 6 enrollments divided by the number of elementary teachers, a binary compliance label defined by a threshold of 35 students per teacher consistent with DepEd policy, and two classifiers — a Random Forest with 100 estimators and an SVM with a linear kernel — trained on a four-feature matrix comprising Enrollees_G1, Enrollees_G6, Teachers_Elementary, and Teachers_JHS, with an 80/20 train-test split and random_state=42.

Both classifiers achieved high predictive accuracy on the held-out test set, and the Random Forest model demonstrated stable generalization performance across 5-fold cross-validation, confirming the viability of this framework as a scalable, reproducible school-level compliance monitoring tool. The detailed classification report confirmed balanced and reliable per-class discrimination for both the compliant and non-compliant classes. Feature importance analysis identified Teachers_Elementary as the dominant predictor of SDG 4.7 compliance by a considerable margin, followed by Enrollees_G1, Enrollees_G6, and Teachers_JHS — a ranking that is both computationally coherent and directly interpretable in institutional terms.

The principal policy implication of these findings is that targeted and equitable elementary teacher deployment is the highest-leverage intervention available to DepEd for improving SDG 4.7 compliance rates across the national public school system. The predictive pipeline developed in this study can be operationalized as an annual compliance screening tool applied to updated DepEd enrollment and employment data at the start of each school year, enabling proactive identification of at-risk schools before classroom overcrowding becomes entrenched.

Future research should extend the feature set to include classroom infrastructure data, school location classifications, regional socioeconomic indicators, and multi-year longitudinal records. Expanding the compliance framework to the Junior High School and Senior High School levels would provide a system-wide view of SDG 4.7 alignment across the full K to 12 continuum. Integration of the classification pipeline into DepEd's Education Management Information System would translate this research into a sustained institutional monitoring capability with direct and measurable impact on educational quality in the Philippines.

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